

Annex II — Relational Automation, Graph Routing, and Gateway Mechanics

BEAP™ / WR Desk™ Canonical Specification — standalone Annex. Version 1.1 (see §II.7). Renumbered from the earlier I.13–I.17 draft to avoid collision with existing Annex I sections. Mapping: I.13 → II.1 · I.14 → II.2 · I.15 → II.3 · I.16 → II.4 · I.17 → II.5. References of the form "Annex I §I.x" point to the existing Annex I — Protocol Enforcement Mechanics.

II.0 Status, Scope, and Relationship to the Canon

This Annex is normative unless explicitly marked as informative.

It defines the deployment scope of BEAP™, the device classes of a deployment, the normative structure of the WR Graph™, the routing and lifecycle semantics of the WR Graph™, the permission-bearing asymmetric handshake and the BEAP™ Gateway, feedback reintegration, the WCR authoring lifecycle, the pairing and sealed provisioning of Field Devices, and Forward Preparation of provisioned content.

This Annex does not redefine or extend wire formats, capsule serialization, cryptographic primitives, transport mechanisms, or PoAE™ execution semantics; these remain exclusively defined in the Canon and its Normative Annex. All enforcement mechanics of Annex I apply unchanged; nothing in this Annex weakens or contradicts them.

This Annex is a normative extension of the Optirando™ Canon within the scope declared above. In the event of conflict regarding wire formats, capsule semantics, cryptographic primitives, transport, or PoAE™ core semantics, the Canon's Normative Annex prevails. In the event of conflict regarding WR Graph™ structure, routing, gateway mechanics, WCR authoring, or Field Device provisioning, this Annex prevails over descriptive text elsewhere in the Canon. The acquisition governance of all WR Graph™ mutations is defined in Annex III (PoAC™); where this Annex classifies an operation under the Annex III mutation taxonomy, that classification is normative.

II.1 Deployment Scope of BEAP™ (Normative)

II.1.1 Current Deployment Scope

BEAP™ is currently specified, implemented, and validated exclusively for deployment within:

- Optirando™ / WR Desk™ environments
- Silicon Cargo environments

Within this scope, all conformance guarantees of the Canon and its Annexes apply in full.

BEAP™ SHALL NOT be represented, marketed, or relied upon as a general-purpose third-party interoperability standard at this stage. Conformance claims by implementations outside the scope above are undefined.

II.1.2 Future Third-Party Opening (Informative)

Opening BEAP™ to third-party implementers is a long-term objective, not a present property of the protocol. Such an opening would require, at minimum:

- a published conformance test suite,
- capsule and gateway certification procedures,
- a governance process for protocol evolution.

Nothing in this Annex implies a timeline or commitment for this opening. Until such a program exists, references to third-party use SHALL be framed as outlook only.

II.1.3 Device Classes and Conformance Scope (Normative)

A conforming deployment distinguishes two device classes:

Host. The trust root of a deployment: the machine or machines operating the orchestrator, the policy engine, the consent store, the vaults, and the authoritative WR Graph™. All authorization, admission, and revocation authority resides with the Host.

Field Device. A mobile or portable device — for example a smartphone or tablet — paired to exactly one Host through the handshake-based device identity system. A Field Device operates exclusively under its Host's trust root. It holds no independent trust root, no authoritative WR Graph™, no policy authority, and no consent authority. It MAY hold sealed prepared packages, replicated read-only graph regions, templates, adapters, and bounded delegated scopes, exclusively as provisioned by its Host under II.6.

Host and Field Device are roles, not machines. A single-node deployment holds both roles on one device, in which case the provisioning constraints of II.6 apply trivially.

The Field Device security model is specified provisionally. Conformance claims for Field Device deployments are limited to the constraints stated in II.6 of this Annex; a complete Field Device enforcement profile — including delegated enforcement tiers and device attestation — is a declared future extension and SHALL be framed as outlook only, consistent with the pattern of II.1.2.

II.2 WR Graph™ Structure, Adapter Activation, and Routing Semantics (Normative)

II.2.1 Adapter Bindings as Graph Content

WR Graph™ nodes and subgraphs MAY carry bindings to fine-tuned, LoRA-class domain adapters as first-class context. An adapter binding associates a context region (device class, workflow stage, topic, publisher relationship) with the parametric competence required to reason within it.

II.2.2 The WR Graph™ as Activation Trigger Layer

The WR Graph™ SHALL act as the authoritative trigger layer for adapter lifecycle operations, namely:

- **Activation** — loading an adapter when its bound context becomes active
- **Hot-swap** — replacing the active adapter when context resolution changes

- **Combination** — composing multiple adapters where a context legitimately spans several bound domains
- **Deactivation** — unloading adapters whose bound context is no longer active

Lifecycle operations SHALL be triggered exclusively by deterministic context-resolution events, including:

- WR Code® scan resolution
- workflow state transitions within an active recorded workflow
- topic classification of the active session against the graph
- handshake scope activation (see II.3)
- **explicit operator selection** — manual activation of an admitted adapter, or selection of a mode carrying adapter bindings; the operator's selection is the deterministic input to the resolution

Operator selection is the strongest form of deterministic resolution, not an exception to it: identical selection against identical graph state yields the identical adapter set (II.2.3). "Exclusively" excludes implicit, probabilistic, self-initiated, or externally triggered activation — an adapter loading itself, a model requesting its own competence, or external traffic causing a load — never operator intent. An operator-selected activation persists for its declared scope (session or mode duration) and is deactivated through the same lifecycle. Operator selection does not bypass admission (II.2.6), does not expand capability scopes, and is logged as a lifecycle event like any other trigger (II.2.3), with the selection recorded as the context reference.

II.2.3 Determinism and Authority Constraints

Note — "capability" in this Annex denotes execution-authorization scope in the sense of the Canon's Capability Token model, distinct from the acquisition-governance sense in which Annex III uses the term.

- Adapter selection SHALL be deterministic: identical context resolution against identical graph state SHALL yield the identical adapter set. Determinism is conditional on graph state: routing state evolves only through mutations governed under PoAC™ (Annex III), so identical resolution against identical graph state remains reproducible across time.
- Adapter activation SHALL NOT expand capability scopes. Capabilities remain exclusively governed by the capability engine, policy engine, and PoAE™ enforcement.
- Every adapter lifecycle event SHALL be logged with context reference, graph version, and adapter identity, and SHALL be auditable.
- Adapter weights SHALL be loaded only within a trust domain of the deploying identity, only after signature and hash verification against their registered derivation or acquisition evidence, and only following admission under II.2.6. The graph selects; an authorized runtime loads. The locality of that runtime — the Host, or a Field Device paired under the Host's trust root (II.1.3, II.6) — is deployment-dependent. Adapter weights SHALL NOT be transmitted to, or loaded by, any device outside the deploying identity's trust domain. Publication or distribution of adapters to external parties is a separate, consent-gated act and is outside the scope of this Annex.

II.2.4 Multidimensional Graph Structure (Normative)

The WR Graph™ is not a conventional knowledge graph. It comprises four interconnected knowledge dimensions over one context space:

1. **Declarative dimension** — the structured knowledge graph itself (entities, relations, constraints, decisions)
2. **Episodic dimension** — long-term interaction memory bound to graph regions
3. **Procedural dimension** — embedded Workflow Context Recorder automations (Annex I §I.6, II.5)
4. **Parametric dimension** — adapter bindings carrying domain competence (II.2.1)

This four-dimensional structure constitutes the normative definition of the WR Graph™ referenced by Annex III (PoAC™) and by descriptive text elsewhere in the Canon.

Background sorting, prioritization, classification, and self-optimization of the four dimensions are performed not by a fifth dimension but by the **graph governance process** — the idle-time optimization machinery specified in Annex III. Every operation of that process is classified under the Annex III mutation taxonomy. References elsewhere in this Annex to background sorting or automatic classification denote this governance process.

Cross-dimension edges enable a single context resolution (e.g. one scan) to select all dimensions simultaneously: what is known, what has happened, which competence applies, and what can be executed.

Adapters carried in the parametric dimension are produced either by the local adaptation pipeline of Annex I §I.12 (project documentation → coverage-gap identification → curated synthetic datasets → iterative local fine-tuning) or acquired externally under the admission requirements of II.2.6; the routing and application of locally produced adapters is demonstrated in the accompanying paper "High-Impact Examples of the Optirando™ System in Action" (§2.3).

II.2.5 Routing Semantics and the Determinism Boundary (Normative)

The WR Graph™'s routing function is preparatory, never executive. It selects and proposes — knowledge regions, adapters, and candidate automation paths — but executes nothing. The determinism guarantee of this specification attaches to the execution artifact, not to the routing:

- A finalized WCR recording, including its tool invocations, SHALL be deterministic: identical recording version plus identical declared inputs SHALL yield the identical action sequence.
- Consequential steps SHALL be gated by user consent at the applicable PoAE™ tier regardless of how the request was routed. No routing or preparation outcome can substitute for, weaken, or pre-satisfy an authorization requirement. Content, context, adapters, or automation paths prepared in advance of need — including sealed prepared packages provisioned to Field Devices (II.6) — remain subject to the applicable PoAE™ tier at the point of consequence and to revalidation as required by their validity constraints.
- Routing state SHALL be derived exclusively from locally recorded, policy-authorized signals (usage, ratings, corrections, authorized decisions). Autonomous adjustment of routing state is limited to reversible, reported operations as classified under Annex III (Class

- 2). External signals — including publisher feedback, incoming capsules, or gateway traffic — SHALL NOT influence routing state.

Consequently, evolution of the routing layer over time affects only which validated options are surfaced, never what an option does when executed or whether it may execute.

II.2.6 Graph Admission Control (Normative)

LoRA-class adapters and WCR recordings SHALL NOT be admitted into the WR Graph™ automatically. Admission requires explicit user consent, consistent with the principle that trust escalation is explicit and never implicit (Annex I §I.7.1). Until admitted, such artifacts remain in their respective libraries or in the sandbox trust domain and are not available to routing.

For artifacts of external origin — including adapters obtained through distribution channels — admission additionally requires verification of acquisition evidence: publisher identity, signature, content hash, version, and declared scope. Admission of an external artifact is a consent-gated mutation (Annex III, Class 3). An external artifact whose acquisition evidence cannot be verified SHALL NOT be admitted; verification failure is fail-closed.

II.2.7 Inspectability and Snapshots (Normative)

The WR Graph™ SHALL be visually inspectable by the user. The system SHALL support taking an immutable snapshot of the graph at any point in time, so that its state remains reviewable even as the graph evolves.

Full graph versioning MAY be implemented; in High-Assurance Mode (Annex I §I.8), graph state relevant to routing and automation admission SHALL be versioned, so that any past routing or admission decision can be reconstructed against the graph state under which it was made.

II.2.8 Graceful Degradation and Domain Boundedness (Normative)

Automations are domain-bound execution logic, not a universal interaction mode. If context resolution finds no matching domain region — including the empty-graph case — the system SHALL degrade gracefully:

- no domain adapter is loaded; the normally selected base model answers,
- no automation path is offered,
- no routing is fabricated from insufficient context.

Absence of domain expertise SHALL be surfaced as such, not masked by a low-confidence match.

II.3 Permission-Bearing Asymmetric Handshake and the BEAP™ Gateway

II.3.1 Extension of Handshake Semantics (Normative)

In addition to counterparty identity, declared purpose, scope constraints, and the handshake identifier (H-ID), a handshake MAY carry automation permission grants. A permission grant SHALL specify:

- the permitted automation classes or named workflow capsules

- per-step capability ceilings (spend, scope, domain restrictions)
- the PoAE™ tier mapping applicable to each consequential step
- validity period and revocation conditions

Permission grants SHALL be recorded in the relational context graph on the user side and in the publisher's backend on the gateway side. Execution SHALL require agreement of both records; any mismatch SHALL fail closed.

Revocation of the handshake SHALL immediately invalidate all future activations under its permission grants, consistent with Annex I §I.4.2. For counterparties, gateways, or Field Devices that are not continuously connected, every permission grant SHALL carry a maximum offline validity period (**lease**). A grant whose lease expires without renewal SHALL fail closed on the disconnected side. Renewal SHALL include a revocation check against the granting record. Revocation therefore takes effect immediately where connectivity exists, and no later than lease expiry where it does not.

II.3.2 BEAP™ Gateway (Normative Definition)

A BEAP™ Gateway is a server-adapted variant of the orchestrator, embeddable into a website and its backend. The Gateway:

- SHALL issue and accept handshake requests on behalf of the publisher
- SHALL connect to the publisher's database exclusively through a bounded, declared connector
- SHALL publish WCR-recorded automations as validated workflow capsules
- MAY maintain a catalog of such capsules covering the publisher's recurring support and configuration topics (e.g. its FAQ corpus)
- SHALL be operable only by publishers that have completed DNS-based publisher verification — the same verification already required for the publication of public BEAP™ capsules

The Gateway inherits all orchestrator-side enforcement obligations; it does not introduce a weaker conformance class.

II.3.3 Registration-Coupled Handshake (Informative)

In the ordinary registration flow of a website, the user MAY additionally conclude an asymmetric handshake with the publisher through the site's BEAP™ Gateway. Registration and handshake remain separable acts: the handshake adds a governed automation relationship on top of the conventional account relationship. The publisher thereby gains a realtime, structured channel to the user's orchestrator instead of static or after-the-fact data exchange.

II.3.4 Asymmetric Channel Model (Normative)

The relationship established under II.3.1–II.3.3 is deliberately asymmetric:

Forward channel (publisher → user) MAY carry:

- workflow capsules (WCR-recorded automations)
- context blocks (per the Relational Context Graph model)
- status and versioning information

Backchannel (user → publisher) SHALL be restricted to:

- PoAE™ authorization records
- explicitly declared data items covered by the permission grant (e.g. address data)
- status and outcome messages
- user feedback artifacts (screenshot plus textual description), subject to text-purity and attachment-handling rules

The backchannel SHALL NOT carry arbitrary content, telemetry, capsules, or undeclared data classes. This restriction SHALL be enforced fail-closed at the user's orchestrator. The asymmetry is a privacy and containment property, not an implementation convenience.

II.3.5 Execution Model and Publisher-Sanctioned Automation (Normative)

Automations delivered over the forward channel are recorded by the publisher against the publisher's own surfaces (website, dashboard, configuration console) and executed in the user's environment against those surfaces.

The decisive governance point: website operators generally do not permit automation on their surfaces. Under a permission-bearing handshake, the publisher explicitly sanctions automation on its own surfaces, and because publisher and surface evolve under one governance, recordings and interface remain aligned (cf. §2.3.3 of the Examples paper).

Where execution occurs on a Field Device, PoAE™ gating authority remains with the Host trust root; local gating on the Field Device is permitted only through a delegated enforcement component explicitly provisioned by the Host and bounded to a declared tier ceiling. Delegated enforcement components are part of the provisional Field Device scope (II.1.3) and are otherwise outside the scope of this Annex.

The following invariants remain unmodified:

- Incoming workflow capsules SHALL traverse the full Ingestor–Validator pipeline (Annex I §I.3) and remain sandboxed until validated.
- Capsules SHALL be cryptographically bound to the H-ID and inherit handshake lifetime constraints (Annex I §I.4.2).
- Recorded automations run to the authorization boundary and no further; every consequential step is gated by its PoAE™ tier as mapped in the permission grant.
- The publisher's Gateway is connected to the publisher's own host over an internal handshake, subject to the same enforcement model.

II.3.6 Capsule Catalogs and Graph-Routed Dispatch (Informative)

A publisher will typically author not one automation but a catalog — for example, on the order of a hundred WCR recordings covering the topics of its FAQ, its onboarding flows, and its recurring configuration tasks. Dispatch of a catalog entry to a user is triggered in one of two ways:

1. **Question-routed dispatch.** The user submits a support request in natural language. The **publisher's server-side graph** performs catalog routing: the question is matched against the catalog's semantic annotations within the Gateway's own graph instance, so that the correct question triggers the correct recording. The Gateway's server-side orchestrator then transmits exactly that capsule to the user over the P2P connection. The user-side WR

Graph™ is neither consulted nor mutated by dispatch; the received capsule affects the user's graph only through the standard admission path (II.2.6) and feedback reintegration (II.4).

2. **Explicit dispatch.** The user triggers a specific automation directly via a button or link on the publisher's website; the corresponding capsule is transmitted without routing.

In both cases the user's orchestrator receives the capsule through the standard ingestion and validation pipeline, executes it locally under the handshake's permission grants, and the user rates the outcome (thumbs up / thumbs down). The rating closes the loop of II.3.7 and, on the publisher side, refines the routing quality of the catalog.

II.3.7 Iterative Support and Feedback Loop (Informative)

Where an automation does not resolve the user's problem, the low-latency P2P relationship enables an iterative loop:

1. Automation runs and fails or terminates incompletely.
2. The user returns a feedback artifact over the backchannel: screenshot plus description ("this did not work").
3. The publisher (or its support tooling) derives a tailored workflow for exactly this state and ships it over the forward channel.
4. The loop iterates until the user confirms resolution ("thumbs up").
5. The resolved session — including the working configuration path — is persisted in the user's WR Graph™, so the graph afterwards holds all details of the successful setup.

Representative use case: fully automated execution of complex third-party configurations (e.g. a Cloudflare setup), where the permission grant covers the configuration steps and any residual failure state is round-tripped as screenshot plus description until a tailored workflow succeeds.

II.4 Feedback Reintegration into the WR Graph™ (Normative)

Outcomes of automation runs, support cases, errors, user actions, and website interactions MAY be reintegrated into the WR Graph™ as versioned context, subject to:

- explicit provenance and temporal ordering (consistent with the append-and-enrich model of Annex I §I.12.2)
- scope binding to the originating relationship, project, or context region
- background sorting by the graph governance process (II.2.4), which assigns reintegrated material to its topic without user effort

Reintegrated feedback SHALL NOT modify capability scopes, permission grants, or enforcement rules; it enriches context only. User feedback MAY, however, cause an automation path to be superseded or deprecated within the graph (e.g. after repeated negative ratings or confirmed interface drift), so that routing ceases to surface stale paths; supersession preserves lineage and is driven exclusively by local, user-authorized signals (II.2.5).

Reintegration operations are classified under Annex III as follows: linking outcomes and feedback events to existing entities is Class 1; supersession or deprecation of an automation path is Class 2 with retained rollback; ingestion of new context sources arising from a session is Class 3.

Where background optimization proposes a revision to an admitted recording — a redundant step removed, a more direct path, an updated constraint — the proposal SHALL NOT replace the active recording silently. Proposed revisions SHALL pass the authoring validation loop (II.5.3) and require explicit user approval; an approved revision replaces its predecessor as an authorized change with recorded rationale and preserved lineage. Over time this yields iterative improvement of the graph per project and per relationship.

II.5 WCR Authoring Lifecycle (Normative)

II.5.1 Authoring Scope and Storage Targets

WCR authoring is not restricted to publishers. Any user MAY author recordings for their own environment; publishers author them for delivery through a BEAP™ Gateway (II.3.6). A confirmed recording is stored in two places:

- the author's recording library, where it is versioned and retrievable as an artifact in its own right, and
- where meaningful, the WR Graph™, embedded into the appropriate domain region as an executable automation path — assignment is performed automatically by the graph governance process (II.2.4), not by manual filing. Automatic assignment is a Class-1 operation under Annex III (derived and reconstructible); the consent for the entity's existence was given by its promotion under II.5.3.

II.5.2 Authoring Paths

A Workflow Context Recorder artifact MAY be authored via two paths:

Path A — Video-derived authoring. The user uploads a video (screen recording, tutorial, or third-party material such as a YouTube video) together with a textual instruction describing the intended workflow. The system derives a candidate semantic manuscript from video and instruction.

Path B — Direct recording. The user performs the workflow while the recorder captures all available signals simultaneously — video, audio, the semantic action sequence, and structural interface metadata — and fuses them into one accurate semantic instruction. No single signal is authoritative; fusion follows the precedence rules of Annex I §I.11.6.

II.5.3 Validation Loop

Authoring SHALL proceed through a staged validation loop:

1. Candidate manuscript is generated (Path A or B).
2. Supervised test run: the candidate is executed in the sandbox; the semantic layer of the execution is recorded and reconciled against the manuscript.
3. Final test run: the reconciled manuscript is executed again for confirmation.
4. User confirmation:
 - **Thumbs up** → the manuscript is promoted to a stored WCR record and automatically classified into the appropriate topic region of the WR Graph™ by the graph governance process (background sorting, no manual filing). Automatic classification is a Class-1 operation under Annex III; the promotion itself constitutes the consent for the entity's existence.

- **Thumbs down** → the user describes what went wrong; the description enters the manuscript as corrective context, and the loop returns to step 2.

Revisions proposed by background optimization (II.4) SHALL pass through this same loop before replacing an active recording.

II.5.4 Preserved Constraints

All constraints of Annex I §I.6 apply unchanged to both authoring paths: no executable macros, no replay scripts, no binding to static selectors or pixel coordinates; the stored artifact is a structured textual manuscript requiring mediated interpretation under BEAP™ governance.

Video-derived candidates (Path A) SHALL be treated as external-origin artifacts and remain confined to the sandbox trust domain until validated (Annex I §§I.2.2, I.3).

II.5.5 Tool Invocation Constraints (Normative)

WCR recordings encompass tool use, but tool use SHALL be treated as an explicit, separately governed element of the recording — not as an implicit side effect of replay:

- Every tool invocation within a recording SHALL be declared in the manuscript with tool identity and declared inputs.
- Tools SHALL be subject to explicit policy whitelisting. Invocation of a non-whitelisted tool SHALL fail closed; whitelisting is a policy-engine decision, never a routing outcome.
- Tool invocations SHALL be routed through the sandbox sub-orchestrator unless a tool is explicitly designated for host execution by policy.
- Tool results SHALL be returned to the calling context in text format only, consistent with the text-purity invariant; binary or opaque results remain confined to the sandbox trust domain.
- Once a recording is finalized, its tool invocations are part of the deterministic execution artifact (II.2.5): fixed, inspectable, and cryptographically bound at the point of consequence.

II.6 Field Device Pairing and Sealed Provisioning (Normative, Provisional)

This section specifies the currently defined constraints for Field Devices (II.1.3). It is deliberately minimal: it states only what a conforming deployment enforces today. The complete Field Device enforcement profile is future work (II.6.6).

II.6.1 Pairing and Trust Root Subordination

A Field Device SHALL be bound to exactly one Host through the handshake-based device identity system. The binding SHALL be revocable by the Host at any time. Upon revocation, all content provisioned to the Field Device under II.6.2 SHALL be treated as expired regardless of remaining lease time, taking effect immediately where connectivity exists and no later than lease expiry where it does not (II.3.1).

A Field Device SHALL NOT establish handshakes, accept capsules, or conclude permission grants with external counterparties in its own right; all such relationships belong to the Host.

II.6.2 Sealed Provisioning

Content delivered to a Field Device — prepared context packages, replicated read-only graph regions, templates, and adapters — SHALL be:

- **sealed to the device identity:** cryptographically bound to the paired device such that it is unusable on any other device,
- **version-stamped:** bound to the WR Graph™ state (graph version or snapshot reference, II.2.7) from which it was prepared, and
- **lease-bound:** carrying a maximum offline validity period under the lease semantics of II.3.1.

Provisioning of non-confidential, already-admitted content to a paired Field Device is a Class-2 operation under Annex III (reversible by recall or expiry, reported). Provisioning of content classified as confidential, of credential references, or of bounded delegated scopes is a Class-3 operation requiring an explicit Consent Record.

II.6.3 Expiry and Revalidation

Sealed content whose lease expires without renewal SHALL become unusable on the Field Device; expiry is fail-closed. Renewal SHALL occur only through contact with the Host and SHALL include a revocation and version check: content prepared against a superseded graph state MAY be renewed only after re-preparation or explicit Host confirmation that the prepared state remains valid.

Nothing held on a Field Device pre-satisfies any authorization requirement (II.2.5). Consequential actions initiated from a Field Device are gated at the applicable PoAE™ tier by the Host, subject to the delegated-enforcement limitation of II.3.5.

II.6.4 Exclusions

The following SHALL NOT be provisioned to, stored on, or delegated to a Field Device under any configuration:

- vault material and raw credentials or secrets (credential references only; resolution occurs within the Host trust domain),
- root identity keys of the deployment,
- the policy engine of record, the consent store, or any consent authority,
- authoritative (writable) WR Graph™ regions,
- revocation authority of any kind.

II.6.5 Forward Preparation (Normative)

Forward Preparation is the assembly of execution-ready content — prepared context packages, templates, adapter bundles, and automation fragments — ahead of need, using idle time and storage to reduce runtime dependence on connectivity and local computation. A prepared package is provisioned content under II.6.2: sealed, version-stamped, and lease-bound.

Forward Preparation has two triggers:

1. **Declared preparation (operator-triggered).** The operator declares an upcoming context — "tomorrow I visit museum X", "Friday: maintenance round on line 3" — and the declaration is the deterministic trigger: the system assembles the relevant context package (graph

subsets, templates, admitted adapters, automation paths for the declared context) and provisions it to the designated device ahead of the declared time. Declared preparation is available on demand and does not depend on the Night Pipeline schedule.

2. **Predictive preparation (system-proposed, opt-in).** Where enabled by configuration, recurring-context patterns in local episodic evidence MAY drive preparation proposals, mirroring the opt-in recurring-action detection discipline of Annex III §5.7: pattern detection over locally recorded events is Class 1; with detection disabled, no such analysis occurs.

Under either trigger, the mutation classification is identical: package generation from admitted content is Class 2 (Annex III §4.5); provisioning follows the classes of II.6.2 (Class 2 for non-confidential admitted content, Class 3 for confidential content, credential references, or delegated scopes).

Constraints, both absolute:

- **No preparation outcome pre-satisfies any authorization requirement** (II.2.5). Prepared trust or policy evaluations are advisory pre-checks at most; every consequential step is gated at its PoAE™ tier at the point of consequence, against the state at that time.
- **Absence of preparation fabricates nothing.** Where no prepared package covers the encountered context, the system degrades per II.2.8: no routing is fabricated, and the absence of prepared context is surfaced as such.

II.6.6 Outlook (Informative)

A complete Field Device enforcement profile — delegated enforcement components with declared tier ceilings, device attestation, offline PoAE™ evidence sealing with reconciliation on reconnect, and a full placement model governing which layers of the architecture may reside on which device class — is a declared future extension. Until such a profile is published, Field Device deployments SHALL rely exclusively on the constraints of this section, and references to broader Field Device autonomy SHALL be framed as outlook only.

II.7 Version History

Version	Date	Change
1.0	2026-07	Initial standalone Annex II: deployment scope, WR Graph™ adapter activation and routing semantics, permission-bearing asymmetric handshake and BEAP™ Gateway, feedback reintegration, WCR authoring lifecycle. Device classes and conformance scope (II.1.3: Host and Field Device as roles; provisional Field Device scope). II.2.4 promoted from informative to normative as the referenced WR Graph™ definition; five layers consolidated to four knowledge dimensions plus the graph governance process (Annex III alignment). II.2.2 trigger list extended with explicit operator selection (manual adapter activation and mode selection) as a deterministic resolution event, with clarification of what "exclusively" excludes. II.2.3 adapter locality generalized to trust-domain formulation with verification-before-load and admission requirements; conditional-determinism clarification. II.2.5 non-pre-satisfaction rule extended from routing outcomes to preparation outcomes; routing-signal wording harmonized with the Annex III mutation taxonomy. II.2.6 extended to external-origin artifacts with acquisition-evidence verification. II.3.1 lease semantics for disconnected counterparties, gateways, and Field Devices. II.3.5 Field Device
1.1	2026-07-09	

Version **Date****Change**

execution placement and delegated-enforcement limitation. II.3.6 routing locus clarified (publisher's server-side graph; user-side graph neither consulted nor mutated by dispatch). II.4 and II.5 reintegration and auto-classification operations classified under the Annex III mutation taxonomy. New II.6: Field Device pairing, sealed provisioning, expiry and revalidation, exclusions; II.6.5 Forward Preparation as a normative mechanism with declared (operator-triggered, on-demand) and predictive (opt-in) triggers, Annex III class mapping, non-pre-satisfaction and no-fabrication constraints; Outlook renumbered to II.6.6. Authority and precedence clause added to II.0.

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